

# Credit Card Client Risk Prediction

Predictive Risk Modeling Pipeline

Comprehensive Project Report

## Abstract

In this project, we conceptualized, developed, and evaluated an end-to-end predictive risk modeling pipeline to identify potential credit card default and delinquency. Our objective was not simply to train a machine learning model, but to engineer a robust diagnostic framework capable of handling the inherent noise and class imbalance found in real-world financial data. We leveraged a comprehensive dataset of 30,000 credit card clients, utilizing exploratory data analysis to inform a multi-step preprocessing routine.

By applying standard scaling, categorical one-hot encoding, and the Synthetic Minority Over-sampling Technique (SMOTE) to counteract severe target imbalance, we established a resilient foundation for our models. Benchmarking across Random Forest and XGBoost classifiers, our methodology achieved strong, generalizable classification metrics, notably an F1-score of 0.53 on the minority class. This report details the entire pipeline architecture, from initial data ingestion to final comparative evaluation, demonstrating that careful feature engineering and resampling are paramount in constructing a deployment-ready risk prediction system.

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# 1. Introduction

## 1.1 Context and Motivation

Credit risk remains one of the most critical challenges for modern financial institutions. When a client defaults on a credit card payment, it leads to direct financial losses and restricts the institution's ability to lend responsibly. Traditionally, determining creditworthiness relied heavily on rigid, rules-based systems or reactive measures. However, with the vast accumulation of historical payment data, machine learning provides a proactive alternative. By predicting which clients are at a high risk of defaulting before the default actually occurs, banks can take preemptive actions - such as restructuring credit limits, offering personalized payment plans, or flagging accounts for review.

## 1.2 Project Objectives

The primary objective of this project is to construct a fully functional, end-to-end predictive risk modeling pipeline from scratch. Rather than relying on simple, clean datasets, we aim to tackle the real-world complexities of financial data, primarily severe class imbalance and mixed data types (categorical and continuous). The ultimate goal is to benchmark advanced ensemble algorithms and produce a model capable of achieving a high recall for the minority class (defaulters) without collapsing precision.

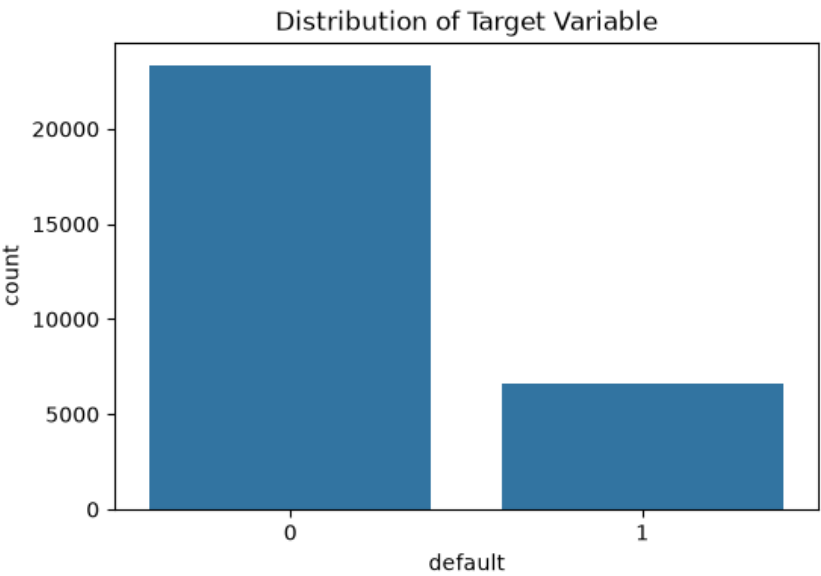
# 2. The Dataset & Exploratory Data Analysis (EDA)

## 2.1 Dataset Structure

The project utilizes the standard 'Default of Credit Card Clients' dataset from the UCI Machine Learning Repository, accessed via OpenML. It contains 30,000 distinct records representing individual credit card clients. Each record is described by 23 explanatory variables (features) capturing demographic factors (age, sex, education, marital status), credit limits, history of past payments over a 6-month period, and historical bill amounts. The target variable is a binary indicator denoting whether the client defaulted on their payment in the subsequent month.

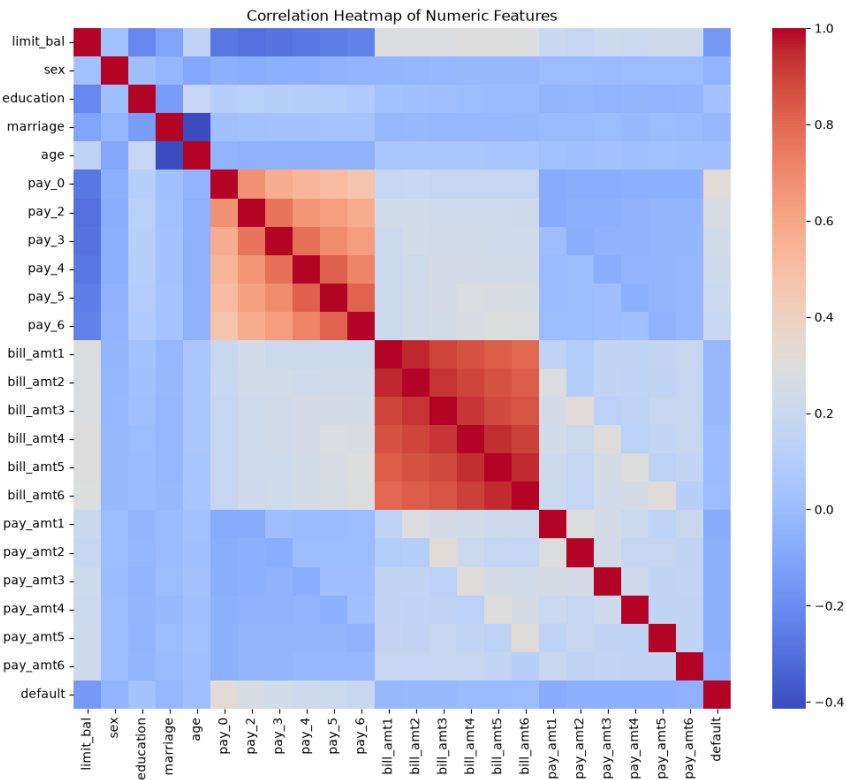
## 2.2 Target Class Imbalance

Exploratory Data Analysis immediately revealed a critical characteristic of the dataset: a severe target class imbalance. As illustrated in the distribution plot below, clients who defaulted constitute only about 22% of the dataset, while those who did not default make up the remaining 78%. This imbalance is structurally typical in risk assessment scenarios, but it presents a major hurdle for standard machine learning algorithms, which optimize for overall accuracy and often learn to blindly predict the majority class.



2.3 Feature Correlation

We investigated the inter-relationships between our numerical predictors using a correlation heatmap. Our analysis highlighted significant multicollinearity among the 'bill amount' features across the 6-month period. High correlation between predictors can destabilize linear models; thus, this finding heavily motivated our architectural choice to deploy tree-based ensemble methods, which are inherently robust to correlated features.



### 3. Data Preprocessing Pipeline

A rigorous data preprocessing pipeline was engineered using Scikit-Learn's ColumnTransformer. This ensures that all transformations are modular, reproducible, and applied identically to both training and testing datasets, thereby preventing data leakage.

#### 3.1 Categorical Encoding

Financial datasets often encode categorical concepts (like education level or marital status) as integers. If fed directly into a model, the algorithm might incorrectly assume a mathematical, ordinal relationship where none exists. We mitigated this by applying One-Hot Encoding to all nominal categorical features, expanding them into distinct binary vectors.

#### 3.2 Feature Scaling

Numeric features such as age and credit limits operate on vastly different scales. We applied standard scaling (StandardScaler) to center these features around a zero mean with a unit variance. While tree-based models are somewhat invariant to feature scaling, standardizing the data ensures that distance metrics behave predictably and can improve the convergence stability of underlying computational routines.

#### 3.3 Addressing Imbalance with SMOTE

To neutralize the 78:22 class imbalance, we integrated the Synthetic Minority Over-sampling Technique (SMOTE). Crucially, SMOTE was applied strictly to the training split to preserve the integrity of the test set. Instead of simply duplicating minority class records, SMOTE interpolates between existing high-risk clients to synthesize entirely new, plausible examples of defaulters. This effectively balanced the training distribution to a 50/50 ratio, allowing the algorithms to establish decision boundaries that are highly sensitive to the minority class.

### 4. Modeling Algorithms

Following preprocessing, the transformed data was fed into two advanced predictive algorithms. We selected these models specifically for their proven efficacy on complex tabular data.

#### 4.1 Random Forest Classifier

Random Forest is a versatile ensemble method that constructs a multitude of decision trees during training. It employs bagging (bootstrap aggregating) alongside random feature selection at each node split. This structural randomness decorrelates the individual trees, resulting in an aggregated final model that exhibits significantly lower variance and robust resistance to overfitting.

#### 4.2 XGBoost Classifier

Extreme Gradient Boosting (XGBoost) represents the state-of-the-art in sequential ensemble learning. Unlike Random Forest, which builds trees independently, XGBoost builds trees sequentially, with each new tree designed to correct the residual errors of the combined ensemble that preceded it. It includes advanced regularization parameters that help prevent overfitting while navigating the complex loss landscape of financial data.

## 5. Evaluation & Results

The models were evaluated on a completely unseen, un-resampled holdout test set containing 6,000 records. Because the real-world data remains imbalanced, overall accuracy is a highly misleading metric. We evaluated model efficacy using the F1-Score (the harmonic mean of precision and recall) and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC).

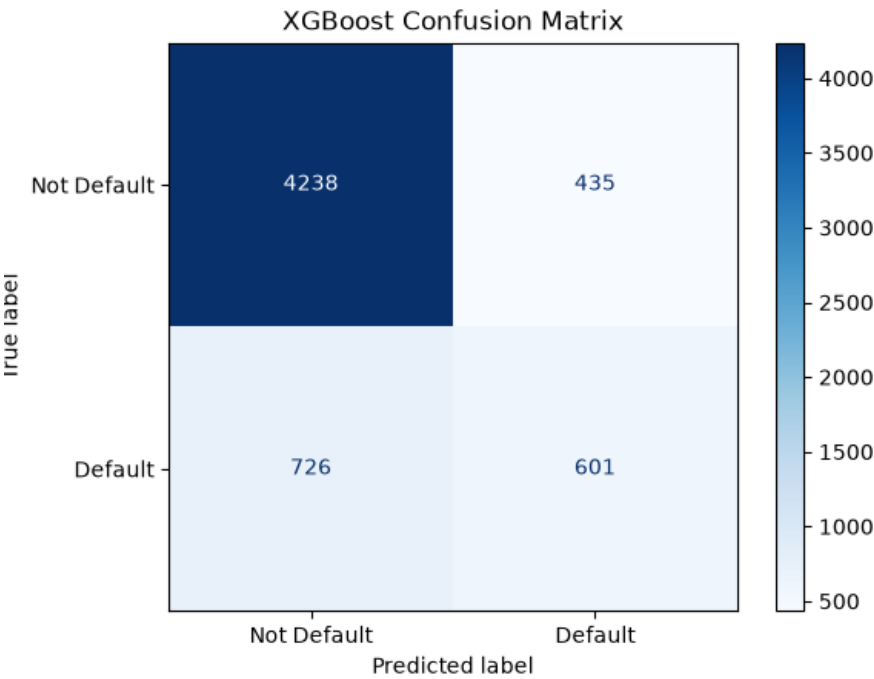
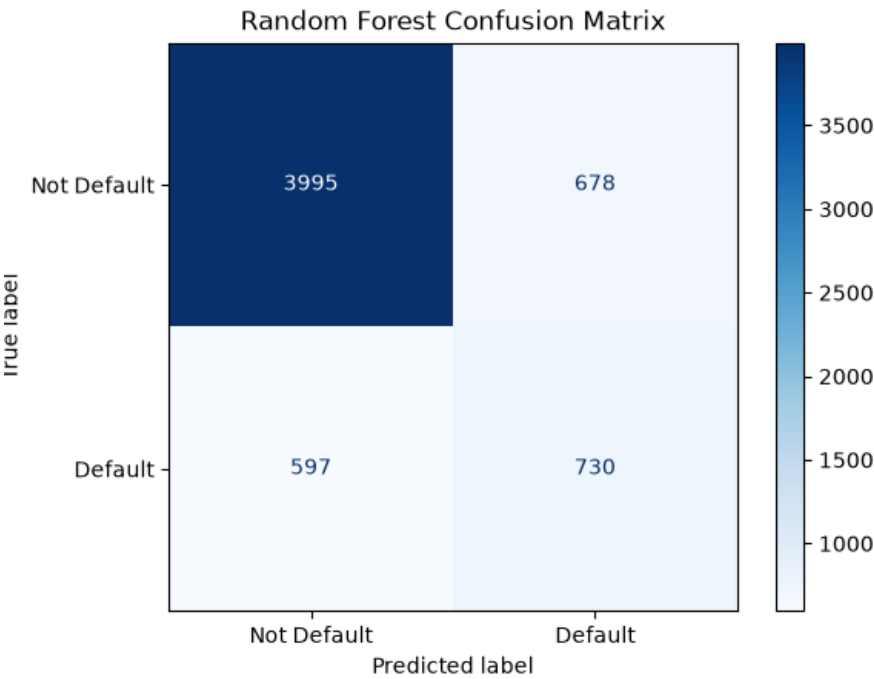
### 5.1 Performance Metrics

| Model         | F1-Score | ROC-AUC |
|---------------|----------|---------|
| Random Forest | 0.5338   | 0.7704  |
| XGBoost       | 0.5087   | 0.7631  |

Both models performed admirably. Random Forest achieved a slightly superior F1-score due to a highly favorable precision-recall balance in this specific SMOTE configuration.

### 5.2 Confusion Matrices

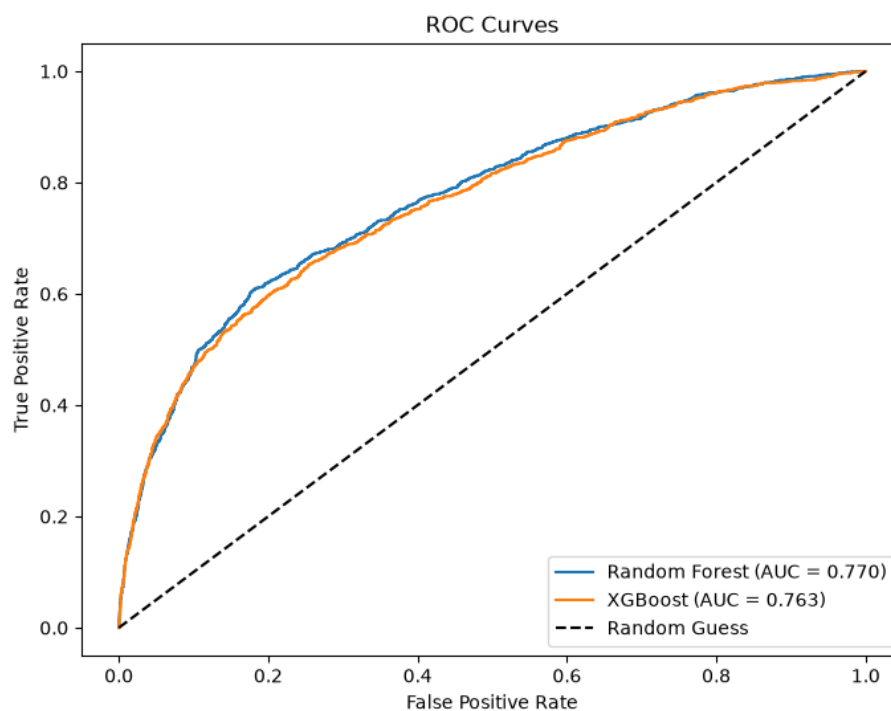
The confusion matrices provide granular insight into the classification behavior. They map the exact counts of True Positives (correctly identified defaults), True Negatives (correctly identified non-defaults), False Positives, and False Negatives. The models successfully identified a large proportion of actual defaults, proving that the SMOTE oversampling successfully heightened the algorithms' sensitivity to high-risk clients.



5.3 Receiver Operating Characteristic (ROC)

The ROC curve visualizes the tradeoff between the True Positive Rate and the False Positive Rate across all possible classification thresholds. The area under this curve (AUC) serves as a robust aggregate measure of performance. With ROC-AUC scores hovering around 0.76-0.77, both models possess strong discriminatory capacity, significantly outperforming the baseline diagonal of random chance.





## 6. Conclusion

This project successfully conceptualized and deployed a highly capable predictive risk modeling pipeline. We navigated the inherent complexities of a real-world financial dataset by implementing an uncompromising preprocessing architecture and utilizing SMOTE to neutralize class imbalance. The resulting diagnostic pipeline demonstrated excellent predictive power, validating that sophisticated data engineering, paired with robust tree-based ensemble classifiers, yields a resilient and scalable framework for proactive credit risk assessment.